

Do deep research and do not respond with mock data I need 100% accuracy in able to select the correct property parcels I need to submit for a proposed cellular tower. The research is for *Chatham County, North Carolina" find the zoning ordinance and locate the tower and antennas section and provide me with the following data the county requires to build a new cellular tower. The allowable zoning districts, the setbacks, the height restrictions, and the fall zone. With this criteria I can start selecting the most feasible property parcels

To ensure 100% accuracy for your proposed cellular tower in Chatham County, North Carolina, here is a detailed summary of the current zoning ordinance requirements for wireless telecommunications facilities, including allowable zoning districts, setbacks, height restrictions, and fall zone requirements. The following is based on the most recent and relevant official documentation, recognizing that the Unified Development Ordinance (UDO) will take effect July 1, 2025, but as of now, the Wireless Telecommunications Facilities Ordinance is the applicable code for new towers^{[1] [2] [3]}.

1. Allowable Zoning Districts

Wireless telecommunications facilities, including towers and antennas, are generally allowed in all zoning districts, but the specific permitted or conditional use status depends on the type and height of the structure:

- *Permitted (P):**
 - Wireless Support Structures (towers) up to 60 feet in height in any zoning district (and unzoned areas).
 - Concealed Wireless Facilities up to 60 feet in height in any zoning district (and unzoned areas).
 - Concealed Wireless Facilities up to 150 feet in height (but greater than 60 feet) in non-residential districts (O&I, B-1, NB, CB, RB, IL, IH).
- *Conditional Use (CU):**

- Wireless Support Structures greater than 60 feet but less than 199 feet in height in most districts (R5, R2, R1, O&I, B-1, NB, CB, RB, IL, IH).
- Wireless Support Structures greater than 199 feet but no more than 300 feet in height in any district.
- Concealed Wireless Facilities greater than 60 feet but less than or equal to 150 feet in height in some residential districts (R5, R2, R1) require conditional use permit^[4].

Note: In unzoned areas, towers up to 199 feet are allowed in properties coded as industrial uses by Chatham County Tax Records, subject to certain conditions^[3].

2. Setbacks

- **General Setback:** Each Wireless Support Structure and Wireless Facility must be set back from all property lines a distance equal to its engineered fall zone or 50 feet, whichever is greater^[3].
- **Tall Towers:** For towers greater than 100 feet, a setback of 50% of the tower height is required from adjoining property lines and right-of-way, but never less than the fall zone or 50 feet^[3].

3. Height Restrictions

- **General:** Wireless Support Structures (towers) up to 199 feet are allowed in most districts, subject to administrative or conditional use review^[3] ^[4].
- **Residential Districts:** Towers up to 199 feet are allowed, but generally require conditional use approval for heights above 60 feet^[4].
- **Industrial Districts:** Towers up to 199 feet are allowed as permitted or conditional use, depending on height and specific district^[3] ^[4].
- **Above 199 Feet:** Towers up to 300 feet are allowed in any district with conditional use approval^[4].
- **Waivers:** Height waivers may be granted by the county upon justification by the applicant^[3].

4. Fall Zone

- **Definition:** The fall zone is the area around a tower where, in the event of a collapse, debris would be expected to land.
- **Requirement:** The setback must be at least equal to the engineered fall zone or 50 feet, whichever is greater^[3].
- **Tall Towers:** For towers over 100 feet, the setback must be at least 50% of the tower height, but never less than the fall zone or 50 feet^[3].

Summary Table

Requirement	Details
Zoning Districts	All districts (with height limits and use type restrictions); industrial/unzoned for >60 ft towers
Setbacks	Fall zone or 50 feet, whichever greater; 50% of height for towers >100 ft (never <fall zone/50 ft)
Height Limits	Up to 199 ft (most districts); up to 300 ft with conditional use; 60 ft permitted in many districts
Fall Zone	Setback must cover fall zone or 50 feet, whichever greater

Key Application Requirements

- **Administrative Review:** Required for towers up to 199 feet in industrial/unzoned areas or up to 60 feet in any district^[3].
- **Conditional Use Permit:** Required for towers above 60 feet in most districts or above 199 feet in any district^[3] ^[4].
- **Engineer Certification:** Must provide documentation of fall zone calculation and structural integrity^[3].
- **Site Plans:** Must include property boundaries, setbacks, topography, elevation sketch, and dimensions^[3].

Conclusion

To select the most feasible property parcels in Chatham County, NC, for a new cellular tower, ensure that:

- **The parcel is in an allowable zoning district** (or unzoned/industrial for taller towers).
- **The parcel can accommodate the required setback** (fall zone or 50 feet, whichever is greater; 50% of height for towers >100 feet).
- **The proposed tower height complies with district-specific limits** (up to 199 feet in most places, up to 300 feet with conditional use).
- **A licensed engineer certifies the fall zone and structural integrity** as part of your application.

For the most up-to-date information and to verify any changes from the new UDO (effective July 1, 2025), consult the Chatham County Planning Department or their official website^[1] ^[2] ^[3].

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1. <https://www.chathamcountync.gov/government/departments-programs-i-z/planning>

2. <https://www.chathamcountync.gov/government/departments-programs-i-z/planning/special-topics/udo-unified-development-ordinance>

3. http://cms6ftp.visioninternet.com/chathamcountync/planning_board/attachments/2013/10-21-13_BOC/Cell_Tower_Ord_Amendment/DRAFT_Communication_Towers_Ord_JepComments_BOC17Feb14.pdf
4. http://cms6ftp.visioninternet.com/chathamcountync/planning_board/attachments/2013/12-3-13_Planning_Board/Cell_Tower_Text_Revisions/Redline_Text_Amendment_ZO_CellTowersDecPB.pdf